


# NPTEL Course on "Computer System Optimization for ML"

Lecture #46

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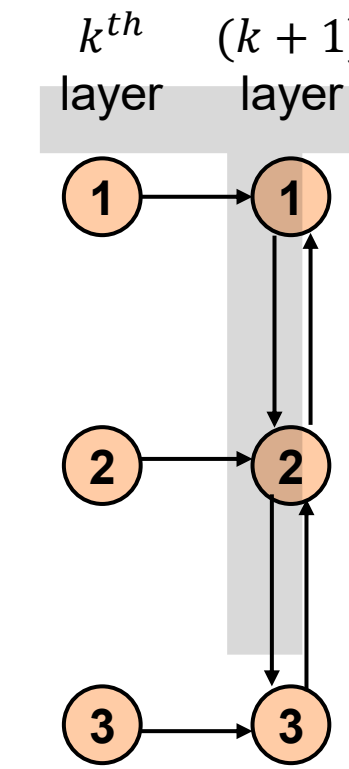
# Agenda

- Minimum Communication Latency (contd.)
- GNN Accelerator with IMC

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# Custom Interconnect

- Two consecutive layers consists of equal number of routers
  - Three routers in this case
- Each router can send/receive only one packet at one cycle
- In one round of communication each routers in $k^{th}$ layer send a packet to each router in $(k+1)^{th}$ layer
- One transaction is finished in 3 cycles
  - Minimum possible latency




(a)

| Link | Cycle | Transaction |
|---|---|---|
| $L_{k,1} - L_{(k+1),1}$ | 1 | $R_{k,1} - R_{(k+1),1}$ |
| $L_{k,2} - L_{(k+1),2}$ | 1 | $R_{k,2} - R_{(k+1),2}$ |
| $L_{k,3} - L_{(k+1),3}$ | 1 | $R_{k,3} - R_{(k+1),3}$ |
| $L_{(k+1),1} - L_{(k+1),2}$ | 2 | $R_{k,1} - R_{(k+1),2}$ |
| $L_{(k+1),2} - L_{(k+1),3}$ | 2 | $R_{k,2} - R_{(k+1),3}$ |
| $L_{(k+1),3} - L_{(k+1),2}$ | 2 | $R_{k,3} - R_{(k+1),2}$ |
| $L_{(k+1),2} - L_{(k+1),1}$ | 2 | $R_{k,2} - R_{(k+1),1}$ |
| $L_{(k+1),2} - L_{(k+1),3}$ | 3 | $R_{k,1} - R_{(k+1),3}$ |
| $L_{(k+1),2} - L_{(k+1),1}$ | 3 | $R_{k,3} - R_{(k+1),1}$ |

(b)

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# Reconfigurable NoC (1)

- **There are two class of CNNs**
  - One is used in edge computing hardware
  - Another is used in cloud-based computing platform
- **We observe similarity in optimal NoC for different networks in each of the two class**
  - Similarity in number of routers for each layer (shown in next slide)
- **On the chip, a number of routers will be reserved for each layer**
- **Before executing the NN algorithm, the proposed algorithm is run (on the chip) to reconfigure the NoC**
  - Deciding the number of routers needed for each layer
  - Deciding the optimal schedule

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# Reconfigurable NoC (2)

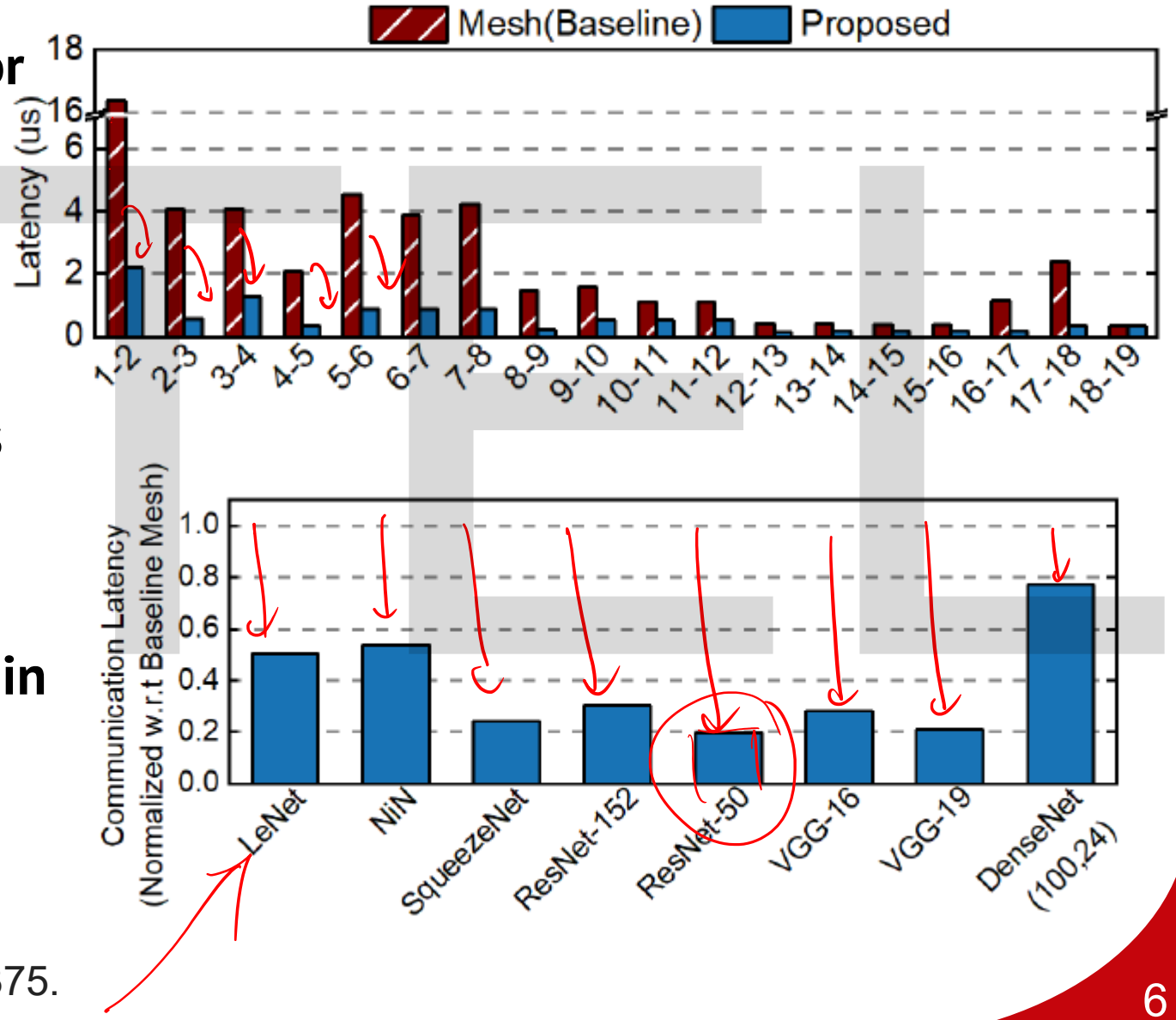
| | lenet-5 | nin-12 | squeeze-26 | vgg-16 | vgg-19 | generic_noc |
|---|---|---|---|---|---|---|
| Layer 1 | 4 | 2 | 5 | 4 | 6 | 6 |
| 2 | 4 | 2 | 5 | 4 | 6 | 6 |
| 3 | 4 | 2 | 4 | 4 | 5 | 5 |
| 4 | 2 | 2 | 6 | 4 | 5 | 6 |
| 5 | | 2 | 6 | 4 | 4 | 6 |
| 6 | | 2 | 4 | 4 | 4 | 4 |
| 7 | | 2 | 4 | 4 | 4 | 4 |
| 8 | | 2 | 4 | 4 | 4 | 4 |
| 9 | | 2 | 3 | 4 | 3 | 4 |
| 10 | | 2 | 4 | 4 | 3 | 4 |
| 11 | | | 4 | 4 | 3 | 4 |
| 12 | | | 3 | 4 | 3 | 4 |
| 13 | | | 3 | 4 | 2 | 4 |
| 14 | | | 3 | 2 | 2 | 3 |
| 15 | | | 2 | 2 | 2 | 2 |
| 16 | | | 3 | 3 | 2 | 3 |
| 17 | | | 3 | | 2 | 3 |
| 18 | | | 2 | | 2 | 2 |
| 19 | | | 3 | | 2 | 3 |
| 20 | | | 3 | | | 3 |
| 21 | | | 2 | | | 2 |
| 22 | | | 3 | | | 3 |
| 23 | | | 3 | | | 3 |
| 24 | | | 2 | | | 2 |
| 25 | | | 3 | | | 3 |
| 26 | | | 3 | | | 3 |

- **Entries in the table shows optimal number of routers required for different networks (proposed approach)**
- **The right-most column (generic_noc) shows number of routers allocated per layer on the chip**
  - Maximum of all networks for that layer
  - The chip can afford all the 5 CNNs shown in the table
- **At runtime, for a given CNN, number of routers required for each layers of the CNN and the corresponding schedules is computed with the proposed technique**
  - Reconfigurability
- **A given network can use only the maximum available routers for the corresponding layer**
  - Can lead to sub-optimality
  - Should still be optimal than mesh-based solution
- **This hardware can support CNNs with up to 26 layers**
- **Similar common features for cloud-computing based NNs**

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# Experimental Results [1]

- Layer-wise improvements shown for VGG-19
  - Up to 8× improvement in communication latency
- Improvement in communication latency is shown for different DNNs
  - Normalized with respect to the baseline communication latency
- Results in up to 25% improvement in inference latency




[1] Mandal, Sumit K., et al. "A latency-optimized reconfigurable NoC for in-memory acceleration of DNNs." *IEEE Journal on Emerging and Selected Topics in Circuits and Systems* 10.3 (2020): 362-375.

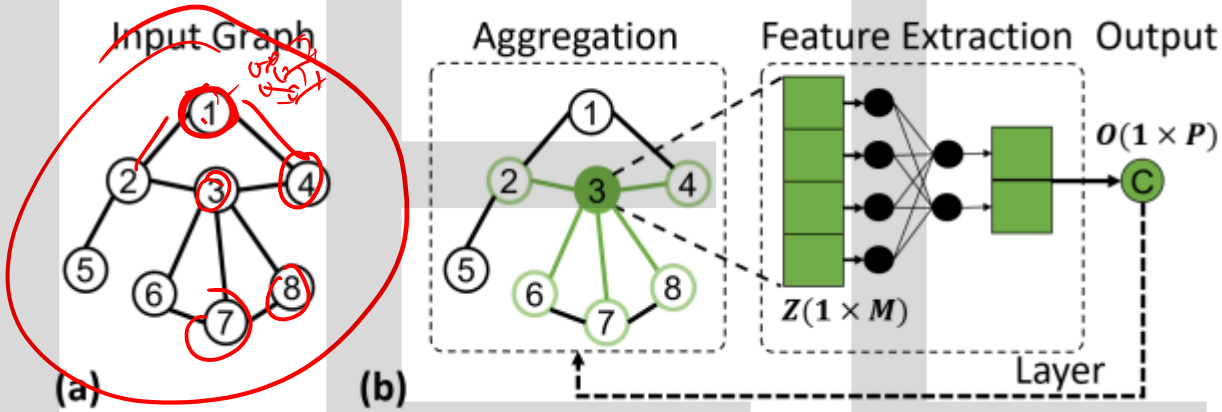
# Agenda

- Recap
- **GNN Accelerator with IMC**

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# Recap on Graph Neural Network (GNN)

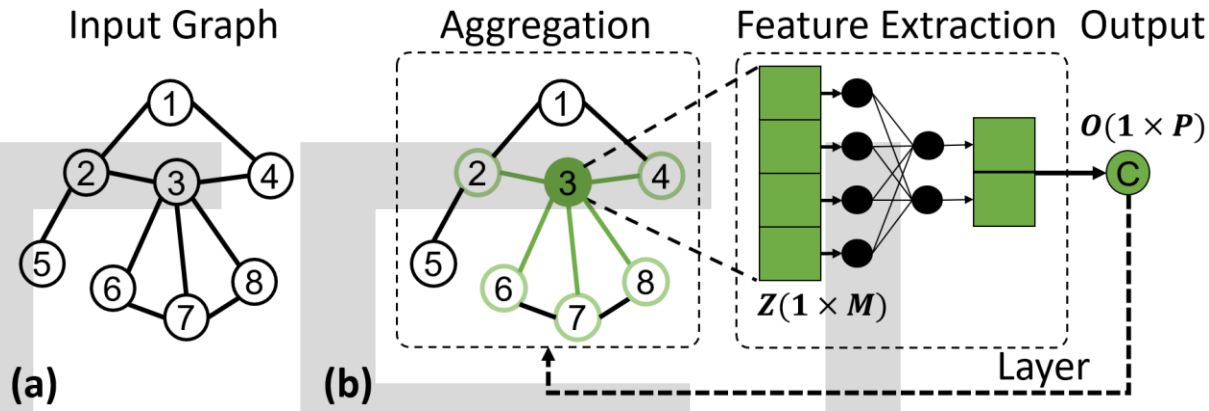
- **Two phases**
  - Aggregation
  - Combination/Feature Extraction
- **Aggregate function aggregates multiple feature vectors from neighbours to form one single feature vector**
  - Different GNNs use different function
- **Combination function transforms the feature vector of each vertex to another feature vector using a MLP**
- **MLP parameters are shared between vertices**

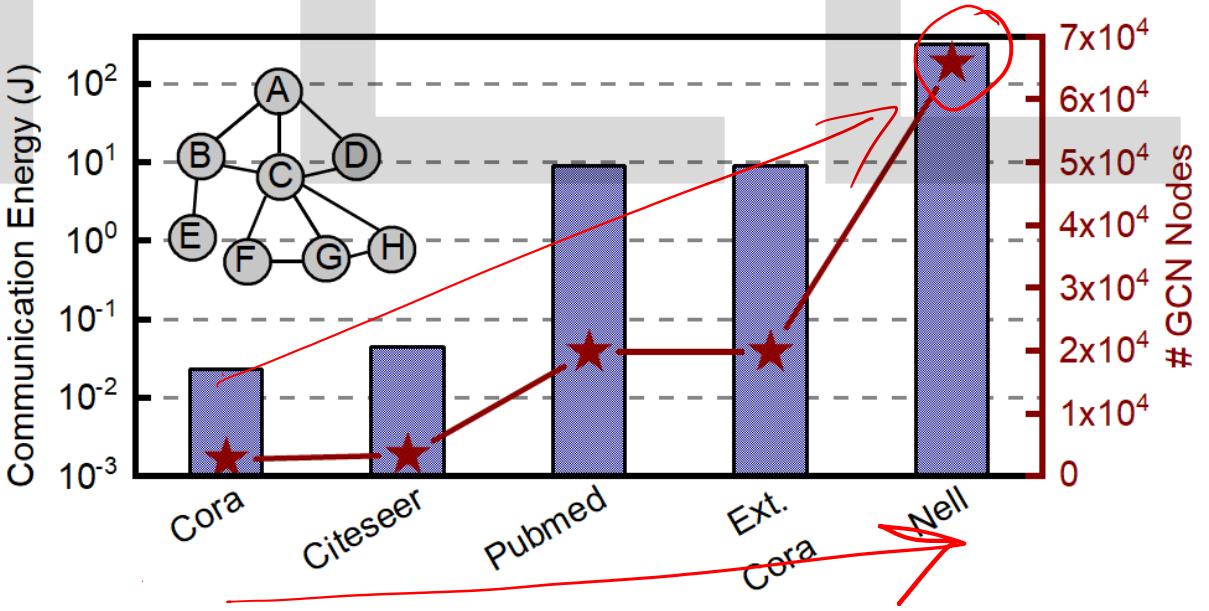



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# Graph Convolutional Network (GCN)

- **Multiple DNNs connected with edges**
- **An example of GCN shown in the top left**
  - Aggregation and combination occur periodically
- **High number of parameters**
  - Nell: 65K nodes, 266K edges
- **A router per node will result in high communication energy**
  - Figure in the bottom right corner

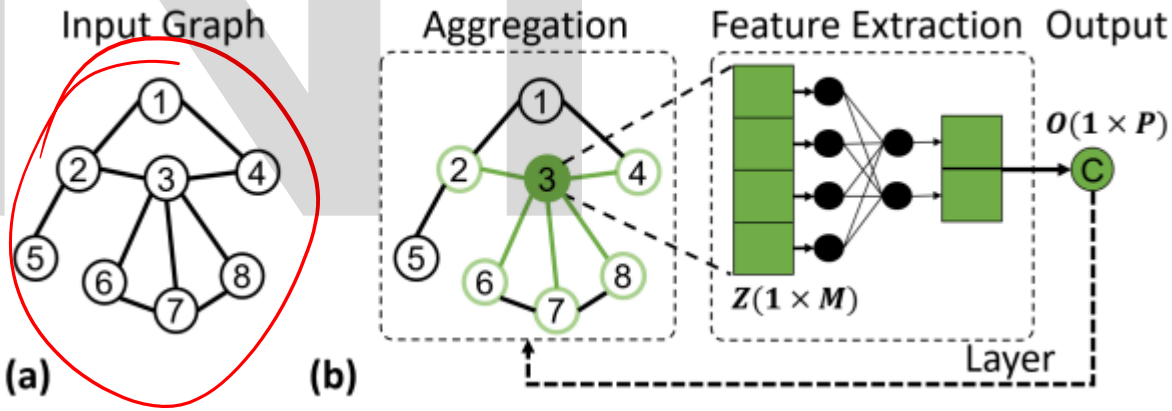



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# GNN Computation: Adjacency Matrix

- **Adjacency Matrix: Representation of the graph**
  - **0 if there is no connection**
  - **1 if there is a connection**
  - **N×N, N is the number of nodes present in the graph**




| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| **1** | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 |
| **2** | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 |
| **3** | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 1 |
| **4** | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| **5** | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| **6** | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| **7** | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 |
| **8** | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 |

8×8

# GNN Computation: Feature Matrix

- A vector of features is present at each node at each iteration/layer
  - Multiplied with Adjacency Matrix to obtain the feature from the neighbors

**Adjacency Matrix**

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| **1** | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 |
| **2** | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 |
| **3** | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 1 |
| **4** | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| **5** | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| **6** | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| **7** | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 |
| **8** | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 |

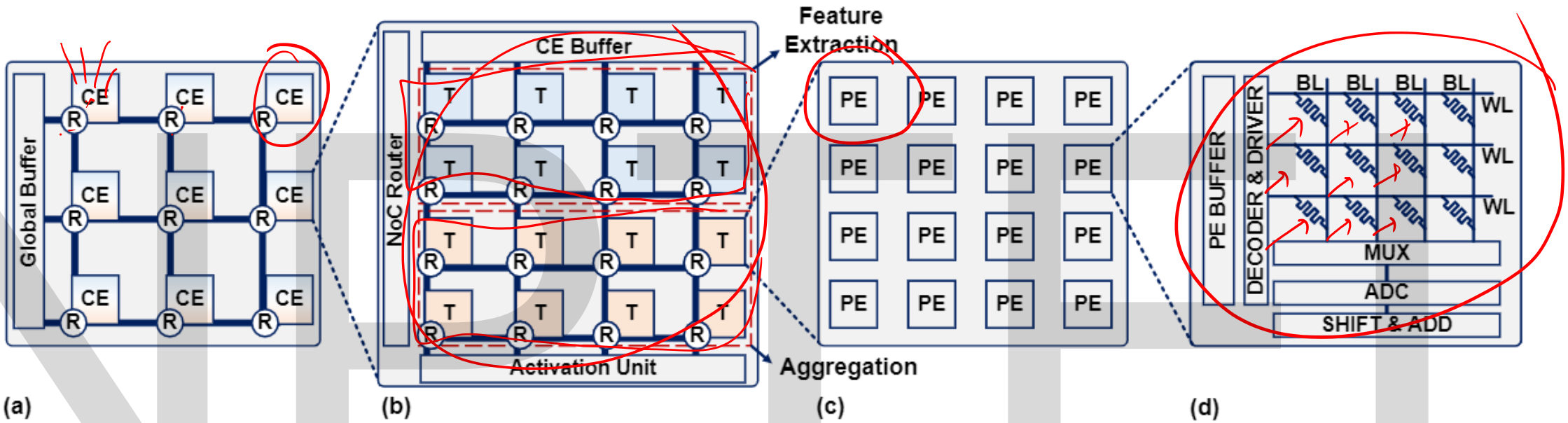
8x8

**Feature Matrix**

| | $f_1$ | $f_2$ | $f_3$ | $f_4$ |
|---|---|---|---|---|
| **1** | 5 | | | |
| **2** | 8 | | | |
| **3** | 3 | | | |
| **4** | 2 | | | |
| **5** | 4 | | | |
| **6** | 9 | | | |
| **7** | 6 | | | |
| **8** | 7 | | | |

8x4 = 8x4

# Communication-aware GCN Accelerator [1]




- **Hierarchical architecture**
- **Several GCN nodes are mapped to a CE**
- **CEs consist of multiple tiles to perform computations for**
  - **Feature extraction**
  - **Aggregation**

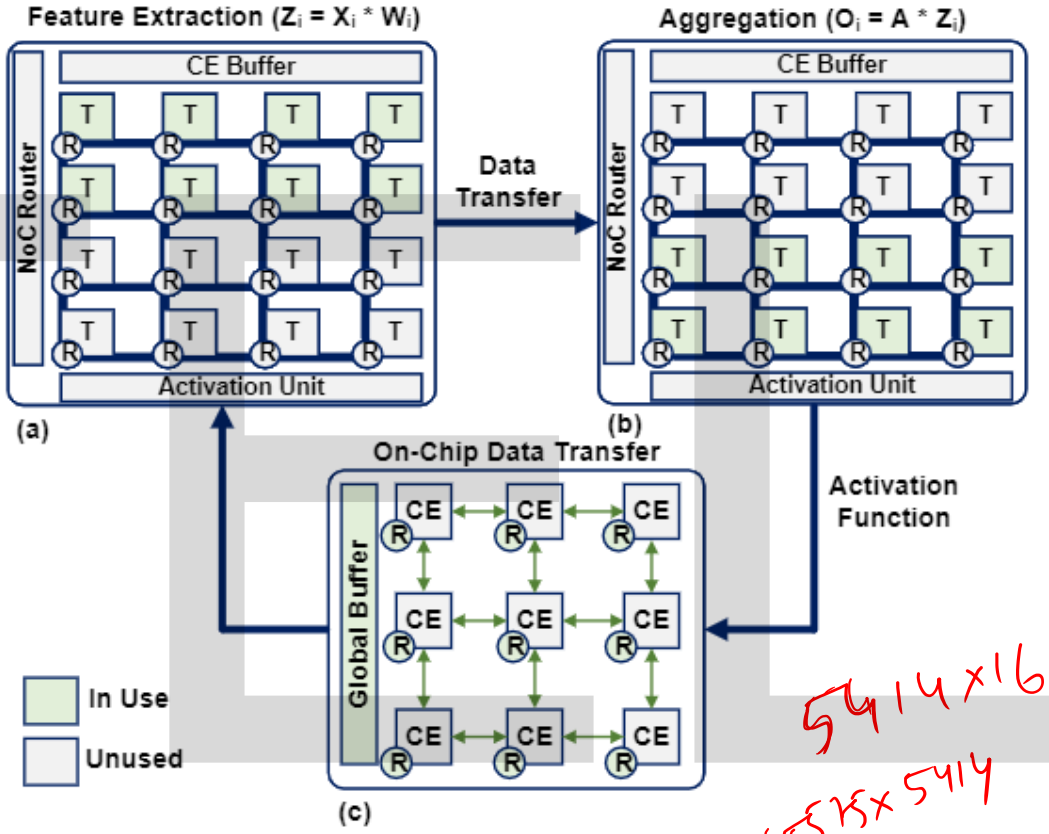

[1] Mandal, Sumit K., et al. "COIN: Communication-aware in-memory acceleration for graph convolutional networks." *IEEE Journal on Emerging and Selected Topics in Circuits and Systems* 12.2 (2022): 472-485.

# Dataflow

- **Feature extraction → Aggregation → On-chip communication**
- **Reduces number of computations**

**Illustration with Nell dataset**

| Matrix | Size |
|---|---|
| A | 65575×65575 |
| X | 65575×5414 |
| W | 5414×16 |




| Type | Multiplication-1 | Multiplication-2 | Total |
|---|---|---|---|
| Aggregation first | 65575 × 65575 × 5414 | 65575 × 5414 × 16 | $2.3 \times 10^{13}$ |
| Feature extraction first (proposed) | 65755 × 5414 × 16 | 65575 × 65575 × 16 | $7.4 \times 10^{10}$ |




# THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)